PROJECT PRESENTATION & TECHNICAL REPORT

AI-Based Movie Recommendation & Ranking System

Semantic matching using Natural Language Processing (NLP), Sentence-BERT embeddings, and Cosine Similarity metrics.

INPUT CORPUS	DEEP MODEL	METRIC	OUTPUT
1M+ Movies Dataset	all-MiniLM-L6-v2	Cosine Similarity	Ranked Top 10 Match

1. SYSTEM WORKFLOW & DATA PIPELINE

The system converts natural-language preferences into high-dimensional semantic vectors and matches them across movie features:

1. **CSV Ingestion:** Ingests comprehensive metadata: Movie Title, Genre, Plot Overview, and Descriptive Keywords.
2. **Feature Fusion & Preprocessing:** Handles missing attributes, normalizes text representations, and creates composite attribute vectors.
3. **Sentence-BERT Encoding:** Transforms text into 384-dimensional dense semantic vectors using transformer architectures.
4. **Cosine Similarity Computation:** Evaluates the directional orientation angle between the user preference vector and the pre-computed catalog embeddings.
5. **Ranking & Persistence:** Normalizes similarity to percentage confidence scores, filters the top 10 items, and exports results to CSV.

Key Mechanism: User natural-language queries bypass rigid keyword matching. Descriptors such as *"space + astronauts + emotional"* correctly align with relevant plot summaries even when exact keyword tokens differ.

2. RECOMMENDATION OUTPUT & EXPERIMENTAL QUERY

Sample Query: *"I want a science fiction movie about space exploration and astronauts"*

Rank	Movie Title	AI Similarity	Rating	Primary Genre
1	The Vivid Reckoning	**67.50%**	4.8 / 10	Sci-Fi / Doc
2	The Burning Labyrinth	**66.91%**	2.7 / 10	Comedy / Sci-Fi
3	The Solemn Empire II	**66.82%**	8.0 / 10	Action / Sci-Fi
4	The Dark Warrior	**66.62%**	2.9 / 10	Drama / Sci-Fi
5	The Forgotten Storm	**66.37%**	2.0 / 10	Thriller / Sci-Fi
6	The Eternal Signal	**65.77%**	4.4 / 10	Documentary
7	The Forgotten Machine	**65.48%**	9.9 / 10	Action / Sci-Fi
8	The Endless Legacy	**65.40%**	3.6 / 10	Thriller / Sci-Fi
9	The Dark Warrior (Edition II)	**65.37%**	3.7 / 10	Comedy / Drama
10	The Forgotten Throne	**65.30%**	8.8 / 10	Action / Drama

3. IMPLEMENTATION ARCHITECTURE (PYTHON / PYTORCH)

```
# Core Vector Embedding & Inference Pipeline from sentence_transformers import SentenceTransformer from sklearn.metrics.pairwise import cosine_similarity # Load Transformer Architecture model = SentenceTransformer("all-MiniLM-L6-v2") # Embed Preprocessed Metadata Strings movie_embeddings = model.encode(df["Combine Text"].tolist(), show_progress_bar=False) def recommend_movies(user_input, top_n=10): user_vec = model.encode([user_input]) similarity = cosine_similarity(user_vec, movie_embeddings)[0] ranked_df = df.copy() ranked_df["similarity_percentage"] = similarity * 100 return ranked_df.sort_values(by="similarity_percentage", ascending=False).head(top_n)
```

The system exports tabular recommendations directly into `movie_recommendations.csv` for production consumption and frontend ingestion.

4. EXPLORATORY DATA ANALYTICS & INSIGHTS

Post-processing visualization pipelines analyze the quality and categorical distributions of top results:

Genre Representation in Output

Categorical distribution across top recommendations:

Documentary	**7 occurrences**
Comedy	**7 occurrences**
Action / Sci-Fi	**4 occurrences**
Thriller	**3 occurrences**
Animation / Drama	**2 occurrences**

Rating vs. AI Similarity Analysis

Observation of rating Independence:

High semantic match scores occur independently of global public ratings (e.g., scores range from 2.0 to 9.9). This confirms the model prioritizes thematic relevancy over popularity bias.

5. CORE AI TECHNOLOGIES UTILIZED

Natural Language Processing (NLP) **Sentence-BERT** **Dense Vector Embeddings** **Cosine Similarity** **Ranking Algorithms** **Data Analytics** **Seaborn & Matplotlib** **Pandas Data Pipeline**

6. CONCLUSION & PRODUCTION USE CASES

This project implements an end-to-end semantic information retrieval system that bridges the gap between conversational human requests and catalog discoverability.

- **Over-The-Top (OTT) Streaming Platforms:** Natural-language recommendation bars (e.g., Netflix, Prime Video).
- **Movie & Media Portals:** Mood-based and thematic exploration engines.
- **E-commerce & Content Matching:** Re-usable semantic search for diverse text-heavy catalogs.